

Unit III
Non-Linear Programming

Course Objective – COB3: To outline various methods for solving non-linear programming problems.

Course Outcome – CO3: To solve non-linear programming problems through standard methods.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1	What is a non-linear programming problem? Non-linear programming problem refers to optimization problem, in which either the objective function or constraints or both, have non-linear terms. It is an extension of linear programming problem.	K1
2	List different types of non-linear programming problem: Some of the standard non-linear programming problems are (i) Unconstrained Nonlinear Programming (ii) Constrained Nonlinear Programming (iii) Quadratic programming (iv) Separable programming (v) Fractional programming	K1
3	What is Bordered Hessian matrix? The bordered Hessian matrix is a tool used to determine whether a given point is a local extremum (maximum or minimum) of a constrained optimization problem, particularly in nonlinear programming. It is derived from the second-order conditions in optimization problems involving equality constraints and its denoted by $[H^B]$	K1
4	When can Lagrangian method be used for solving non-linear programming problems? The Lagrangian method can be used in those non-linear problems in which objective function is non-linear whereas all constraints are linear (i.e.) Maximize or Minimize $z = f(x_1, x_2, \dots, x_n)$ subject to, $G_i(x_1, x_2, \dots, x_n) = b_i, \quad i = 1, 2, \dots, m$ $x_1, x_2, \dots, x_n \geq 0$ where f alone has non-linear terms and all G_i 's are linear.	K1
5	Mention the conditions by which an objective function can be maximized while using Lagrangian method: The stationary point will give maximum value to objective function if the sign of each of the last $n - m$ principal minors of the bordered Hessian matrix is the same as that of $(-1)^{m+1}$	K1
6	Mention the conditions by which an objective function can be minimized while using Lagrangian method: The stationary point will give maximum value to objective function if the sign of each of the last $n - m$ principal minors of the bordered Hessian matrix is the same as that of $(-1)^m$	K1

7	What are Kuhn-Tucker conditions? Kuhn-Tucker conditions are used to obtain stationary points for a function with nonzero constraints. (i) $\varphi_j \geq 0, j = 1, 2, 3, \dots, m$ (ii) $\frac{\partial L}{\partial x_i} = 0, i = 1, 2, 3, \dots, n$ (iii) $\varphi_j g_j(x_1, x_2, x_3, \dots, x_n) = 0, j = 1, 2, 3, \dots, m$ (iv) $g_j(x_1, x_2, x_3, \dots, x_n) \leq 0, j = 1, 2, 3, \dots, m$	K1
8	What is a quadratic programming problem? A quadratic programming problem is an optimization problem where the objective function is a quadratic function and all the constraints are linear in nature.	K1
9	Write the standard form of quadratic programming problem: Maximize or Minimize $z = \sum_{i=1}^n c_i x_i + \sum_{i=1}^n \sum_{j=1}^n c_{ij} x_i x_j$ subject to, $G_i(x_1, x_2, \dots, x_n) = b_i, i = 1, 2, \dots, m$ $x_1, x_2, \dots, x_n \geq 0$ where all G_i's are linear.	K1
10	Name some methods that can be used for solving quadratic programming problems: Quadratic programming problems can be solved using (i) Wolfe's method (ii) Beale's method	K1
11	Differentiate between linear and non-linear programming. Linear programming optimizes a problem where the objective function and all constraints are linear. In Nonlinear programming, the objective function may be nonlinear, but the set of constraints may be linear or nonlinear.	K1
12	State any two methods used to solve non-linear programming problems. 1) Lagrangian method 2) Quadratic programming	K1
13	Define Lagrange's multiplier method. It is a mathematical technique used to find the local maximums or minimums of a multivariable function when the input is subject to specific constraints.	K1
14	What is the role of Hessian matrix in Lagrangian method? Hessian matrix assesses the local curvature of the Lagrangian function to determine if a constrained critical point is a local minimum, local maximum or a saddle point.	K1
15	State any two applications of non-linear programming in engineering. 1) Engineers use NLP for the real time optimization and design of chemical plants. 2) In Portfolio management, maximize returns while managing risk.	K1

Part B

S.No.	Question	BTL*
1	Solve the following nonlinear programming using Lagrangian method Optimize $z = 2x_1^2 - 3x_2^2 + 18x_2$ subject to $2x_1 + x_2 = 8$ $x_1, x_2 \geq 0 \quad (16)$	K4

2	Solve the following nonlinear programming using Lagrangian method Optimize $z = x_1^2 + 2x_2^2 + x_3^2$ subject to $2x_1 + x_2 + 2x_3 = 30$ $x_1, x_2, x_3 \geq 0 \quad (16)$	K4
3	Solve the following nonlinear programming using Lagrangian method Optimize $z = 4x_1 - 0.02x_1^2 + x_2 - 0.02x_2^2$ subject to $x_1 + 2x_2 - 120 = 0$ $x_1, x_2 \geq 0 \quad (16)$	K4
4	Discuss the working procedure & Kuhn-Tucker conditions in detail and hence solve the following nonlinear programming problem Maximize $z = 3x_1^2 + 14x_1x_2 - 8x_2^2$ subject to $3x_1 + 6x_2 \leq 72$ $x_1, x_2 \geq 0 \quad (16)$	K4
5	Discuss the working procedure & Kuhn-Tucker conditions in detail and hence solve the following nonlinear programming problem Maximize $z = x_1^2 + x_1x_2 - 2x_2^2$ subject to $4x_1 + 2x_2 \leq 24$ $5x_1 + 10x_2 \leq 30$ $x_1, x_2 \geq 0 \quad (16)$	K4
6	Discuss the working procedure & Kuhn-Tucker conditions in detail and hence solve the following nonlinear programming problem Maximize $z = x_1^2 + x_1x_2 - 2x_2^2$ subject to $4x_1 + 2x_2 \leq 24$ $x_1, x_2 \geq 0 \quad (16)$	K4
7	Discuss the working procedure & Kuhn-Tucker conditions in detail and hence solve the following nonlinear programming problem Maximize $z = 6x_1^2 + 5x_2^2$ subject to $x_1 + 5x_2 \geq 3$ $x_1, x_2 \geq 0 \quad (16)$	K4
8	Write the working rule for solving a quadratic programming problem using Wolfe method. Formulate a two-phase simplex method for the following quadratic programming problem using Wolfe method. Maximize $z = 6x_1 + 3x_2 - 2x_1^2 - 3x_2^2 - 4x_1x_2$ subject to $x_1 + x_2 \leq 1$ $2x_1 + 3x_2 \leq 4$ $x_1, x_2 \geq 0 \quad (16)$	K4
9	Write the working rule for solving a quadratic programming problem using Wolfe method. Formulate a two-phase simplex method for the following quadratic programming problem using Wolfe method. Maximize $z = 40x_1 + 6x_2 - 2x_1^2 - 8x_2^2 - 4x_1x_2$ subject to $6x_1 + 2x_2 \leq 36$ $x_1, x_2 \geq 0 \quad (16)$	K4